Classical Field Theory — Exercise Sheet 7

Modes, complex pairs and admissibility

Hints and answers

Use $\{F,G\} = \int (F_\phi G_\pi - F_\pi G_\phi)$ and $\Omega = \int \delta\phi \wedge \delta\pi$. Except for the explicitly finite torus model in Problem 1, functionals have Lecture 7's smooth Schwartz-valued gradients. A formal distributional derivative does not automatically belong to that class.

Exercise 7.1 (Real oscillators in a finite torus model). On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, fix an integer $N \geq 2$ and use the real basis

$$e_0 = (2\pi)^{-1/2}, \qquad e_{n,c} = \pi^{-1/2}\cos(nx), \qquad e_{n,s} = \pi^{-1/2}\sin(nx), \quad 1 \leq n \leq N.$$

Let $\phi = q_0 e_0 + \sum_{n=1}^{N}(q_{n,c}e_{n,c} + q_{n,s}e_{n,s})$ and let π have the same expansion with coefficients p.

(a) Verify orthonormality and compute the pulled-back canonical form and the Hamiltonian $H = \frac{1}{2}\int_{\mathbb{T}}(\pi^2 + \phi_x^2 + m^2\phi^2)\mathrm{d}x$, where $m \geq 0$. Derive the oscillator equations and treat the zero mode at $m = 0$.

(b) Define $\widehat{\phi}_n = (2\pi)^{-1/2}\int e^{-inx}\phi\mathrm{d}x$, and similarly $\widehat{\pi}_n$, for $|n| \leq N$. Express them in the real coordinates and compute their complex bilinear brackets, including $n = 0$.

(c) Compute all nonzero real coefficients and the energy of the data $\phi = 3\cos(2x) - 2\sin x$, $\pi = 5\sin(2x) + \cos x$.

Hint. Integrate products of sines and cosines before reading off the canonical coordinates. The complex Fourier coefficient contains both a cosine and a sine coordinate; reality relates the opposite modes.

Answer. The form is $\delta q_0 \wedge \delta p_0 + \sum_{n,\alpha=c,s}\delta q_{n,\alpha} \wedge \delta p_{n,\alpha}$. Each frequency is $\sqrt{n^2+m^2}$ and $\{\widehat{\phi}_n, \widehat{\pi}_l\} = \delta_{n+l,0}$. The given data have energy $\pi(33 + 13m^2/2)$.

Exercise 7.2 (A complex scalar is two real canonical pairs). Write $\Phi = \phi_1 + i\phi_2$, with both real fields Schwartz, and take

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\nabla\Phi|^2 - m^2|\Phi|^2\big), \qquad m \geq 0.$$

Derive the two momenta and Hamiltonian by real variation. Compute all functional derivatives of $Q = \int(\phi_1\pi_2 - \phi_2\pi_1)\mathrm{d}^d x$ and its action on both fields and both momenta. Identify its finite flow, prove $\{Q,H\} = 0$, and verify admissibility in the product of the two real Schwartz phase spaces. Evaluate Q on $(\phi_1,\phi_2;\pi_1,\pi_2) = (f, 2f; -f, 3f)$, where $f \in \mathcal{S}(\mathbb{R}^d;\mathbb{R})$ and $\|f\|_2 = 1$.

Hint. First expand the absolute squares into their two real terms. The charge must rotate the momentum components as well as the field components. For conservation, integrate the two Laplacian terms by parts.

Answer. $\pi_a = \partial_t\phi_a$ and $H = \frac{1}{2}\sum_{a=1}^{2}\int(\pi_a^2 + |\nabla\phi_a|^2 + m^2\phi_a^2)$. The flow multiplies both Φ and $\pi_1 + i\pi_2$ by $e^{i\alpha}$; $\{Q,H\} = 0$. The given data have $Q = 5$.

Exercise 7.3 (A smooth window and a sharp interval). In one dimension let $w \in C_c^\infty(\mathbb{R};\mathbb{R})$ be a fixed window and set

$$F_w[\phi,\pi] = \frac{1}{2}\int_{\mathbb{R}} w(x)\phi_x(x)^2 \mathrm{d}x, \qquad F_\#[\phi,\pi] = \frac{1}{2}\int_0^1 \phi_x(x)^2 \mathrm{d}x.$$

Compute both derivatives against arbitrary real Schwartz variations. Prove F_w is admissible. Is $F_\#$ an admissible functional on the whole Schwartz phase space? Test the field $\phi(x) = (x+x^2)\eta(x)$, where $\eta \in C_c^\infty(\mathbb{R})$ equals one on a neighbourhood of $[0,1]$. Display its endpoint distribution terms.

Hint. Do not discard the two interval endpoint terms. To rule out a smooth representing gradient, test it with bounded bumps shrinking to one endpoint.

Answer. $\operatorname{grad} F_w = (-\partial_x(w\phi_x), 0)$ is Schwartz. The sharp derivative is $-\mathbf{1}_{(0,1)}\phi_{xx} + \phi_x(1)\delta_1 - \phi_x(0)\delta_0$. For the specified field it is $-2\mathbf{1}_{(0,1)} + 3\delta_1 - \delta_0$, so $F_\#$ is not admissible on the whole phase space.

Exercise 7.4 (The sign of the Hamiltonian commutator). For admissible F, G, let $H_F = D(\operatorname{grad} F)$ and $\mathbb{J}(a,b) = (b,-a)$ as in Lecture 7. Define the vector-field bracket by

$$[X,Y](z) = DY(z)X(z) - DX(z)Y(z),$$

the sign for which it acts on functionals as $X(YK) - Y(XK)$. Use the actual bracket-gradient formula of Lecture 7 to prove $X_{\{F,G\}} = -[X_F, X_G]$. For a nonzero example, choose real $f \in \mathcal{S}$ with $\|f\|_2 = 1$, put $q = \int f\phi$, $p = \int f\pi$, and take $F = q^2/2$, $G = p^2/2$. Compute both vector fields, their commutator, and $X_{\{F,G\}}$.

Hint. Differentiate $X_F = \mathbb{J}\operatorname{grad} F$. In the example the smeared coordinates satisfy $\{q,p\} = 1$, so the commutator must be nonzero.

Answer. $\operatorname{grad}\{F,G\} = H_F\mathbb{J}\operatorname{grad} G - H_G\mathbb{J}\operatorname{grad} F$ gives the minus sign. In the example $[X_F, X_G] = (-qf, pf)$, whereas $X_{\{F,G\}} = (qf, -pf)$.